AaS! 281: What Makes the Scattered Disk So Wild?

Podcast: Ask a spaceman

AI Executive Summary

1. Core Thesis This episode uses an Old West frontier metaphor to explain the Scattered Disk, a region of the solar system situated between the Kuiper Belt and the Oort Cloud. The core argument is that despite simulations predicting Neptune's gravitational migration over 4.5 billion years should have ejected all outer objects, the Scattered Disk persists as a dynamic, unstable population. The discussion categorizes these persistent objects into four groups (Deputies, Settlers, Drifters, Ghosts) and highlights significant gaps in understanding regarding their mass loss, origin mechanisms beyond simple scattering, and the limitations of current observational capabilities.

2. Key Arguments & Data

3. **Solar System Geography:** The transcript maps the solar system as a civilization progression: Inner Planets (Civilized Cities), Asteroid Belt (Stockyards), Jupiter/Saturn (Frontier Towns), Uranus/Neptune (Last Outposts of Law). The Scattered Disk is defined as the "Badlands" or empty desert beyond these frontiers.
4. **Orbital Dynamics:** Objects in the Scattered Disk have perihelia near Neptune’s orbit (30–50 AU) but aphelia extending 100 to 1,000+ AU. Orbits are wildly eccentric and inclined (up to 40 degrees), unlike the orderly Kuiper Belt or spherical Oort Cloud.
5. **The Stability Paradox:** Simulations indicate Neptune should have ejected all scattered disk objects into interstellar space or captured them by inner giants over 4 billion years, yet tens of thousands remain in place. The total mass is estimated at roughly 10% of Earth's mass.
6. **Origin Model (Nice Model):** Developed in mid-2000s at the Observatoire de la Côte d'Azur, this model posits that Jupiter, Saturn, Uranus, and Neptune migrated outward from closer to the sun. Neptune ("the Sheriff") cleared planetesimals, scattering some into resonant orbits and ejecting others.
7. **Four Sub-populations:** The episode divides Scattered Disk objects into:

    - **Deputies:** Resonant objects (e.g., Pluto, Orcus) locked in orbital ratios with Neptune that prevent close encounters (2:3, 1:2 resonances).
    - **Settlers:** Detached objects (e.g., 2004 XR90 "Buffy") with perihelia too far from Neptune to be affected by its gravity.
    - **Drifters (Centaurs):** Orbits cross the paths of giant planets (between Jupiter and Neptune). They are unstable (< few million year lifespans) and feed inner solar system comets (e.g., Halley's Comet history).
    - **Ghosts (Sedenoids):** Highly detached objects (Perihelion > 50–76 AU, e.g., Sedna). Fewer than a dozen known; origin potentially linked to stellar flybys or capture rather than local formation.
8. **Mass & Discovery Discrepancy:** Directly found members (~300) vastly underrepresent total population (estimates of tens of thousands >100km, hundreds of thousands smaller). Initial solar system disk models suggest 30–50 Earth masses in material; current mass is <0.1 Earth masses (99% ejected/lost).

9. **Observational Future:** The Rubin Observatory's first light in 2025 is expected to increase known Scattered Disk population by a factor of ~10, but boundaries between Kuiper Belt, Scattered Disk, and Oort Cloud remain contested.

10. Actionable Takeaways

11. **Monitor the "New Frontier":** Track discoveries from the Rubin Observatory (VSOP) beginning in 2025 to see how new data will resolve boundary definitions and mass estimates of the Scattered Disk.
12. **Refine Solar System Models:** Use this episode's classification of Deputies, Settlers, Drifters, and Ghosts as a framework for analyzing future exoplanetary system dynamics involving gravitational scattering.
13. **Understand Stability Limits:** Recognize that even stable-looking orbits in the outer solar system are subject to long-term instability due to stellar flybys or galactic tides rather than just planetary interactions.

Full Transcript

Okay, so I decided that the theme of this episode would be the Old West, you know, dusty saloons, frontier towns, bandits on the horizon. And I was considering doing the entire episode in an old-timey western-saddle-up partner condo way. And I did it for more than two sentences and came to the conclusion that it was exhausting, both for me and for you and so normal Paul Sutter voice, but the theme sticks around.

Most of our mental maps of the solar system stop at Neptune. Sometimes we may include Pluto as a usual solar system goes past that, isn't it? And if we're especially nerdy, we also know about the Kuiper Belt or Kuiper Belt, however you want to say it, which includes Pluto and some other friends as a sort of outskirts. And way, way, way out there is the Oort Cloud, which is that ever so ambiguous cloud where the long period comets come from. But there's another place, somewhere in between, past the last outpost of civilization but before the true wilds of the Oort cloud and interstellar space. It's a place of drifters, outlaws, and the aforementioned bandits. The badlands of the solar system. Now to make our journey out to the badlands, we have to start with the familiar and work our way out. We have the inner planets, Mercury, Venus, Earth, Mars. These are anywhere from 0.4 to 1.5 AU. AU is astronomical unit, the average distance between the Earth and the Sun. These are the settled cities, the streetlights, the cobblestones, the carriages, the opera houses. I don't know why that came to mind, but it spoke to me of civilization. I'll pass.

The settled cities at two to four AU is the asteroid belt, the dusty stockyard, you know, where all the all the farms and cattle sit around. And then we have Jupiter and Saturn at five and 10 AU respectively. These are the frontier towns. They're big. They're loud. They're the centers of the attention there. There were the railroad stops way past them. At 19 and 30 AU, we have Uranus and Neptune. These are the very last outposts of the law. And everything past Neptune is the outer frontier. Now, the outer frontier itself has different parts, starting with the edges, which are just past the last town on the frontier and the last railroad junction. There's the Kuiper Belt, the Homestead region. This is where where settlers are just starting to make

Farms for themselves. This is a 30 to 50 AU from the Sun. It's this donut around Neptune's orbit. It's mostly icy, rocky-ish bodies left over from the formation of the solar system. Orbits are mostly circular, mostly in the plane of the planets, mostly well-behaved, mostly orderly. There are, of course, exceptions. Pluto lives here. So does Makemake.

Haumea, Kaur, Orcus, they're past civilization, yes, but still they largely follow the rules. And then way on the other side is the Oort cloud, which is 2000 to 200,000 AU. We're talking up to a light year out from the sun possibly at its greatest extent. It's a roughly spherical shell, so orbits are all over the place. This is the reservoir of the long period comets. It's never been directly imaged. Its existence is only inferred from where the long period comets come from. It is the empty desert beyond the solar system. No one has ever set foot out here. No one has ever seen this horizon. It's deduced to exist because things occasionally wander in from that desert.

But we don't care about those today. We care about what's in between. Past the homesteaders, but before the desert wastes. These are where the outlaws live, separate from civilization, but still tied to it. And we'll talk about them more after this commercial break. Welcome back. What sits between the homesteaders and the desert wastes? is the scattered disk. It sits between the Kuiper Belt and the Ork Cloud, but it kind of overlaps with both. It's this gray zone in between. It has no sharp boundary. It's definitions are fuzzy and constantly argued about as definitions tend to be. For us, though, a rough working idea of what the scattered disk is that these are objects with perihelion, which means it's their closest approach to the sun.

is near Neptune's orbit. So we're talking the closest they get to the Sun is 30 to 50 AU out, but they're far this distance. Their appellian is way out. It's 100, 500, sometimes over a thousand AU. So their orbits are wildly eccentric, wildly inclined. Some are tilted 30 degrees, 40 degrees even more from the plane of the planets. So they're not. as orderly as the Kuiper Belt itself, but they're not as wild as the Oort Cloud. They kinda come close to the outer edges of the solar system, but then they wander really far out. It's this in between. No two objects of the scattered disk are quite alike. Each is on its own wild trajectory. There are Probably tens of thousands of members of the scattered disk that are larger than 100 kilometers across,

though we've only directly found a few hundred so far, so we're guessing based on limited samples here. Total mass is roughly the same as the Kuiper Belt, maybe more, maybe a little less. So a decent amount of mass, you know, like 10% ish of a total mass of the earth scattered all around it.

well, scattered disk. And now you see why it has the name scattered disk because these objects are scattered all over the place. And it's lonely out here. There are no fences, there are no grids, there are no settlers, trails run in every direction, some go up, some go down, some circle back for a century, then vanish over the horizon for a thousand years. From our perspective here in the settled parts of the solar system, every so often on the horizon, you can just make out a silhouette. Might be a drifter. Might be one of those bandits. Might be a ghost. Almost certainly a chunk of ice. That just happens to graze into the farthest edge of our horizon before disappearing and sometimes never to be seen again. The distances out here are so vast that the individual members of the scattered disk hardly ever interact with each other. They don't talk to each other. They don't even know anyone else is out there.

The spaces are just too vast. The distances are too great. It's all loners out here. There are no collisions. There are no mergers. There are very few gravitational interactions amongst the members themselves. And what's most remarkable about the scattered disk is that, yeah, it's full of comets and other sorted chunks of rubble, but it's also home to worlds, planets, well dwarf planets, technically, but we won't get into that now. There's like Aris, which is roughly the size of Pluto. Its orbit is nearly sideways compared to the solar system, but it's it's there There's gong gong a thousand kilometers across named for a Chinese water god by the way, they're Sedna Never even comes near Neptune closest it gets a 76 AU out far this it gets is 900 plus a you we don't even have a firm number on its Orbit, that's how hard it is to observe Sedna exists more in rumors than in hard data, but these are worlds We didn't even know about the scattered disk until the mid 1990s This is new territory that we're just now beginning to explore and it's hard because these objects are small They're dim and they're far away, which is the axis of evil when it comes to astronomers You know give me at least one of those and I can get good data if you have all three it's just hard. And the more we learn about the scattered disk, the more confusing it becomes. And

there's one major problem we've discovered about the scattered disk, one thing that really, really bugs us. Just like any resident of the badlands beyond the frontiers of civilization, nothing is stable out in the scattered disk. Any object whose parahealing closest approach is near Neptune will over millions to billions of years get perturbed.

by Neptune during those close passes. Because sometimes when one of these objects gets kinda close to the solar system, Neptune is on the other side of the solar system so there's no big deal. But then give it enough orbits over enough time over millions of years, every once in a while when one of these scattered disk objects comes close, Neptune is nearby and Neptune is big. Neptune dominates the dynamics of the outer solar system because it's the only big thing out there. And so slowly, over the course of time, Neptune's gravity can alter what these objects do. And here's where the problem emerges. Over the age of the solar system, you have foreign change billion years, those little gravitational interactions should have accumulated and either flung these objects all the way out into interstellar space.

or dropped their perihelions down to where the other giant planets can grab it and then eject it. That's what gravitational perturbations do. Every time there's an interaction with Neptune, there's a chance. And the chance builds up with every single interaction. The orbit becomes more and more unstable and either it just gets flicked out of the solar system right away or tossed down into the inner solar system where it's then you're encountering Saturn, Jupiter, And those guys do not play around. They do not like interlopers. They do not like random objects from the scattered disk. They don't like bandits showing up in town. Simulation after simulation says that the scattered disk should have been cleared out billions of years ago, and yet it's not. It's right there. Tens of thousands of objects are still out there on the wildest orbits in the solar system, defying every single model of what should have happened. So...

How did the scatter disk get there in the first place and what keeps replenishing it? The first question is a little easier to answer than the second. Believe it or not, I think in some previous episode recorded in the dim and distant past of this podcast, I actually referred to Neptune as the

sheriff of the outer solar system. So it means either that the Badlands frontier metaphor is especially a rich ground or I'm just embarrassingly uncreated. But anyway, Neptune is the sheriff of the outer solar system, the last representative of law and gravitational order. But Neptune hasn't always been where it is. The giant planets did not form where they currently sit. They form closer into the sun and they migrated outward over the first few hundred million years. Neptune specifically formed somewhere around 15 to 20 AU. Now it's all the way out at 30.

That's a hefty haul for a planet that's 17 times the mass of the Earth. The framework that explains this is called the Nice model, which was developed at the Observatoire de la Côte d'Eure and Nice France in the mid-2000s. So that's nice. And there are other models of the formation of the solar system. The Nice model is the dominant one. It actually doesn't matter. Basically, every version of the outer solar system's early history involves having the large planets form closer in and then migrate outwards, especially Neptune. And when the solar system was forming beyond the giant planets, which were just getting their act together, there was a thick disk of leftover stuff of planetesimals, these icy bodies ranging all the way up from Dust to Pluto, totaling, I don't know, 2050 Earth masses, all told. And when the solar system first formed, all these planetesimals, all these proto-Earths, their orbits were mostly circular, mostly in the ecliptic plane, they were well-behaved, and they didn't know what was coming. And then something happened. There was some sort of instability between Jupiter, Saturn, Uranus, and Neptune. Almost all models of the formation of the solar system agree.

Some even have a fifth planet that formed along with them and then got ejected in the process. But no matter what, Neptune gets flown outwards, and it just eats its way through the planetesimal disk. Every close encounter transfers energy and angular momentum between Neptune and the planetesimal. The sheriff is on the horseback, riding out from town, clearing out the bandits as he goes, and either objects get flung outward out of the solar system altogether or inwards towards the sun, which is the possible origin of the leap. heavy bombardment. Some get trapped in orbits that resonate with Neptune, like Pluto. Like if you're gonna stick around, you gotta follow the orders of the sheriff. And so that's how we get a scattered

disk. Some objects live right on the edge where they get kicked, but not enough to be completely ejected from the solar system. They get just enough energy.

Their orbits get twisted just enough, wildly eccentric, these long, looping ellipses that bring them close to Neptune, but then really far from Neptune, just a little bit more energy and they'd be gone, where their ellipses would stretch all the way out into interstellar space and they'd free themselves from the grips of the Sun's gravitational influence altogether. These were objects that were scattered, but into a disk-like shape in the solar system, so it's the scattered disk. See, astronomy is easy. Patreon.com slash PM Sutter is how you can learn more astronomy goodness like the origins of the term scattered disk. That's Patreon.com slash PM Sutter, and I'm truly grateful for all of your contributions. So this neat and tidy story has a problem though. It explains how the scattered disk formed. It was planetesimals, most of whom got completely ejected and some just got a little mixed up there. But Neptune is still there. And any outlaws that get too close to the sheriff do get ejected. So 4.5 billion years later, how do you have a mechanism that is able to populate the scattered disk, but also keep it there for a really long time? Well, it turns out there's probably not just one neat and tidy answer. And that's because the Scatterdisk, the Badlands of the Solar System, is impopulated by just one kind of object. It's not just a bunch of icy bits left over from the formation of the Solar System, it's actually community of objects. And each one has their own particular origin, story, and behavior. And when woven together gives us the full picture of the Badlands. And so when we ask, how could this population possibly survive, we need to look into who's actually in.

the scouter disk, and then each one gets its own little tail. And we have four kinds of survivors out here in the Badlands. These four kinds I call the deputies, the settlers, the drifters, and the ghosts. The deputies are the resident objects. These are objects locked in resonance with Neptune. Their orbital period is a simple ratio of Neptunes, like two to three, or one to two, three to five, four to seven, so that every say three orbits of this object, Neptune gets five. Or every two orbits of this object, Neptune gets one. What this means is that even though the object crosses Neptune's orbital distance, the resonance ensures that it always makes the crossing when

Neptune is on the other side of the sun. So the object and the sheriff never meet. By the way, the two to three resonance are called the Plutinos.

Because that's what Pluto does. The one to two objects are called the two Tinos. Not making this up, folks. Anyway, what keeps these objects in line is that the resonance is self-correcting. If you nudge the object slightly, its next orbit brings it back to a safe geometry. It's dynamically stable on billion-year time scales. And that's why I like to think of these as the deputies, because even though they and the sheriff never meet, They still obey the orbital mechanics of Neptune. Pluto's one, Orcus is another, Ixion is another, thousands of smaller ones. It's the resonance that keeps them in line. Neptune isn't picking them off because they're behaving. They're always staying on the opposite side of the solar system when they get close. And then there are the settlers. These are the detached objects. These are objects with high eccentricity. They look scattered, but their closest approach is actually pretty far away from Neptune. We have a few of them. There's 2004 XR90, which is nicknamed Buffy, by the way, 2015 KG163, which does not have a cute nickname and many others. So they're members of the scattered disk.

But their closest approach doesn't even bring them close to Neptune, close enough to matter where Neptune's gravity is going to affect them. Because Neptune can't perturb them, their orbits are essentially stable indefinitely. They sit out there quietly following orbits that take them hundreds of AU out and back, never running into anyone. It's not exactly clear how they became so detached without getting kicked out of the solar system because if it was just Neptune doing the work, Then their closest approach should still be near Neptune, so something had to lift them out of their present orbits. We have some candidates that are all speculative. Maybe a passing star during the solar system's youth tugged on these objects just enough, got rid of most of them, but a few just plucked into slightly higher orbits and left them alone. Maybe they're galactic tides doing the same thing.

Maybe there are encounters with other planets that no longer exist. Maybe there are some other interactions we have yet to identify. These are settlers who are well beyond city limits, well beyond the protection of the law. Know what they're doing is a little bit wild, but haven't quite left way

beyond the horizon yet. That takes us to the Drifters. These are the centaurs. These are objects on orbits that cross the giant planets that actually end up between Jupiter and Neptune. They're not technically in the scattered disk anymore, but they're from the scattered disk. They're named for, you know, mythological creatures. These are half planet, half comet. Individually, they're named after individual centaurs. Chiron, Cariclo, Folis, Nessus. They're unstable.

They'll only live a few million years, way shorter than the age of the solar system. Which means every centaur we see today that is hanging out amongst the giant planets was recently delivered from farther out. And where do they come from? They come from the scattered disk. It's a slow leak. Every so often, Neptune finally gets a good gravitational shot at a scattered disk object and drops its perihelion inward. It drops its orbit inward. and then it becomes a centaur. Where do they go? Well, you guessed it. Either they get flung out of the solar system entirely by Jupiter, or they spiral inward and become a short period comet. Haley's comet was once a centaur, was once a member of the scattered disk. It got pulled in by Neptune, juggled around a little bit by Jupiter, and then became a regular short period comet.

Which means once upon a time every short period comet we encounter here in the inner solar system was once a resident of the Badlands and it's proved that this scattered disc is still leaking. And the idea is, hopefully, that this leakage is slow enough to not drain the entire population by now, but that's an open question. And the last are the ghosts, the sedenoids. These are objects so distant and so detached that they don't behave like anything else in the outer solar system. Their perihelion is way beyond Neptune, greater than 50 AU, sometimes greater than 75 AU. Their apelion is hundreds to thousands of AU. They're named after Sedna, the first one found. The total known population is fewer than a dozen. Almost certainly, there's a much larger population hidden out there.

But they're so faint and they're so far out that we can only catch them near perihelion briefly once every 10,000 years or so. And we have the same problem with these ghosts, the sedenoids, as we did with the settlers, the detached objects. They're so far out from the solar system that nothing in

the solar system could have put them that far out. So what did? I don't know. Maybe, again, close passes by other young stars. Maybe stellar flybombs. Maybe these are objects captured by other star systems entirely. Maybe the Sedenoids are aliens born around another sun and then caught by ours billions of years ago. Nobody agrees on where they came from. Nobody agrees on how many they are. Some of them may not even be from around here. The closest they get is on our most distant horizon and then they disappear forever. They'll never be seen again. I don't know about you, but that's To me, that's the definition of a ghost. That's our four member populations of the scattered disk. And all of them show that the disk isn't static, it's alive, it's an active dynamical machine that's running right now. Most of its members aren't stable enough orbits to last for billions of years. And what is unstable is so barely unstable that the leak is slow enough that even now, billions of years later, a good chunk of it still remains.

But the truth is we barely understand the scattered disk. We've directly cataloged maybe a few hundred objects. Population estimates suggest tens of thousands larger than 100 kilometers, but in hundreds of thousands smaller, but there are so many open mysteries. Where did all the stuff go? Simulations of the early solar system show that the early disk started with 30 to 50 Earth masses in material. Now there's less than 0.1 Earth masses. Where did... 99% of it go, the answer is probably ejected. But, yeah, the details are contested. Nobody knows where the Senoids came from. Is it a stellar cluster, passing star, captured aliens, something we haven't thought of? Every candidate explanation has problems. There's a problem with the inclination of their orbits, the angle relative to the plane of the solar system. Scattered-ist objects have a much broader range of orbital tilts than Neptune scattering alone, predicts.

If Neptune was just responsible for their positions, we should see a lot less tilt to their orbits. Something in the history of the solar system produced a wilder population than the simplest models allow. There's the question of the boundary. Where does the scattered disk end in the Oort cloud begin? Where does it end in the Kuiper Belt begin? It's contested. It's contested. It's contested. Are some members of the Scatterdisk not even native to our own solar system? Are they captured from another star? We don't know. The Varice Rubin Observatory, which had its first light in 2025, and its first

surveys are now beginning to yield data, is expected to increase the known population of Scatterdisk objects by roughly a factor of 10. So it's still a ridiculously embarrassingly tiny population of Scatterdisk objects, but it's 10 times more than we knew before.

And I can't tell you what we're gonna find. Every year, we're finding new and new more and more objects in the Scattered Desk. They're weirder with every passing year. The map of the Badlands is being redrawn constantly. And it will be a long, long time before this frontier ever closes. Thank you to Patrick H for the question that led to today's episode. Thank you to all your questions. It's askaspaceman at gmail.com or the website askaspaceman.com. Please keep sending me questions. Please drop a review on your favorite podcasting platform. It really helps the show visibility. And of course, thank you to all my top Patreon contributors. That's patreon.com slash PM Sutter. I'd like to thank my top contributors this month. They're Justin G, Chris L, Bertha M, Cory D, Michael P, Naila, Sam R, Joshua, Scott M, Rob H, Scott M, Louis M.

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